Simplifying Radical Expressions

Algebra Worksheet · Grade 8–10

Name: _____ Date: _____

Learning Objectives

- Simplify square roots by factoring out perfect squares
- Simplify cube roots by factoring out perfect cubes, including negative radicands
- Add, subtract, and multiply radical expressions by combining like radicals

Problems

1. Simplify the square root below:

$\sqrt{36}$

2. Simplify the square root below:

$\sqrt{50}$

3. Simplify the square root below:

$\sqrt{72}$

4. Simplify the cube root below:

$\sqrt[3]{-54}$

5. Simplify the cube root below:

$\sqrt[3]{128}$

6. Add the radical expressions below:

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$-5\sqrt{3} - 3\sqrt{3}$

7. Simplify and then add the radical expressions below:

$\sqrt{45} + \sqrt{20}$

8. Multiply the radical expressions below and simplify:

$\sqrt{6} \times \sqrt{2}$

9. Simplify the cube root expression below:

$\sqrt[3]{27m^3n^5}$

10. Simplify and then subtract the radical expressions below:

$3\sqrt{98} - 2\sqrt{50} + \sqrt{8}$

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Simplifying Radical Expressions — Answer Key

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Answer Key

1. Answer: 6

- 36 is a perfect square because 6 × 6 = 36
- So $\sqrt{36} = 6$

2. Answer: $5\sqrt{2}$

- Factor 50 into a perfect square factor: 50 = 25 × 2
- $\sqrt{50} = \sqrt{25} \times \sqrt{2} = 5\sqrt{2}$

3. Answer: $6\sqrt{2}$

- Factor 72 into a perfect square factor: 72 = 36 × 2
- $\sqrt{72} = \sqrt{36} \times \sqrt{2} = 6\sqrt{2}$

4. Answer: $-3\sqrt[3]{2}$

- Factor 54 into a perfect cube factor: 54 = 27 × 2
- $\sqrt[3]{(-54)} = \sqrt[3]{(-27)} \times \sqrt[3]{2} = -3\sqrt[3]{2}$

5. Answer: $4\sqrt[3]{2}$

- Factor 128 into a perfect cube factor: 128 = 64 × 2
- $\sqrt[3]{128} = \sqrt[3]{64} \times \sqrt[3]{2} = 4\sqrt[3]{2}$

6. Answer: $-8\sqrt{3}$

- Both terms have the same radical $\sqrt{3}$, so combine the coefficients
- –5 – 3 = –8, giving $-8\sqrt{3}$

7. Answer: $5\sqrt{5}$

- Simplify $\sqrt{45} = \sqrt{(9\times5)} = 3\sqrt{5}$
- Simplify $\sqrt{20} = \sqrt{(4\times5)} = 2\sqrt{5}$
- Add like radicals: $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$

8. Answer: $2\sqrt{3}$

- Multiply inside the radical: $\sqrt{6} \times \sqrt{2} = \sqrt{12}$
- Factor $\sqrt{12} = \sqrt{(4\times3)} = \sqrt{4} \times \sqrt{3} = 2\sqrt{3}$

9. Answer: $3mn\sqrt[3]{(n^2)}$

- Factor: $\sqrt[3]{27} = 3$ (perfect cube), $\sqrt[3]{(m^3)} = m$ (perfect cube)
- For n^5, write $n^5 = n^3 \times n^2$, so $\sqrt[3]{(n^3 \times n^2)} = n\sqrt[3]{(n^2)}$
- Combine: $3 \times m \times n \times \sqrt[3]{(n^2)} = 3mn\sqrt[3]{(n^2)}$

10. Answer: $13\sqrt{2}$

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- Simplify $2\sqrt{50} = 2\sqrt{(25\times2)} = 2\times5\sqrt{2} = 10\sqrt{2}$
- Simplify $\sqrt{8} = \sqrt{(4\times2)} = 2\sqrt{2}$
- Combine: $21\sqrt{2} - 10\sqrt{2} + 2\sqrt{2} = 13\sqrt{2}$

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